

Shuaipeng (Frank) Dong

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[Personal Website](#) · [Google Scholar](#) · [LinkedIn](#) · [GitHub](#)

RESEARCH INTERESTS

My research interests include computer vision, continual learning, and knowledge distillation, with a focus on class-incremental visual recognition and memory-efficient lifelong learning for image classification and object detection.

EDUCATION

The Pennsylvania State University

University Park, PA

B.S. in Computer Science · B.S. in Mathematics (Systems Analysis)

May 2023

Cumulative GPA: 3.55 / 4.00 · Dean's List (multiple semesters) · College Honors Student

PUBLICATIONS & PATENTS

Conference Publications

- [1] **S. Dong** and D. Whitefield, "Video Bandwidth Prediction by GEO Satellite Terminals for QoE," *41st International Communications Satellite Systems Conference (ICSSC)*, vol. 2024, pp. 26–31, 2024.
- [2] **S. Dong**, "Predicting Invasive Ductal Carcinoma Using a Deep Convolutional Neural Network," in *Proc. 3rd International Conference on Artificial Intelligence and Education (ICAIE)*, pp. 191–196, 2022.

Patents

- [3] **S. Dong**, "Bandwidth Prediction Using Machine Learning," U.S. Patent Application No. 19/081,355, filed Mar. 2025 (*pending*).

RESEARCH EXPERIENCE

Knowledge Distillation for Class-Incremental Object Detection

2026 – Present

Independent Research · Computer Vision / Continual Learning

- Studying catastrophic forgetting in class-incremental object detection on COCO, where detectors must learn new categories sequentially while retaining performance on previously learned classes.
- Reproducing and benchmarking continual-learning and incremental-detection baselines, including LwF, iCaRL, ERD, and YOLO-IOD, to analyze degradation of old-class performance under sequential class addition.
- Designing teacher–student schemes that distill logit- and feature-level knowledge from a frozen prior-task detector, testing the hypothesis that response and feature distillation preserve old-class accuracy without large exemplar memories.
- Rebuilding the YOLO-World open-vocabulary detection backbone with newer-generation YOLO architectures to test whether stronger feature extractors improve detection performance on COCO under the incremental protocol.
- Redesigning the distance computation at the core of the distillation objective—replacing the classic L2 distance and KL divergence between teacher and student outputs with alternative distance measures, via redesigned loss functions or direct metric substitution—to ease catastrophic forgetting and overfitting.
- Quantifying retention and plasticity under standard incremental-detection protocols on COCO—per-step and average mAP, forgetting, and backward/forward transfer; manuscript in preparation.

Deep Convolutional Neural Networks for Medical Image Classification

2022

Undergraduate Research · The Pennsylvania State University

- Investigated whether a parameter-efficient convolutional design could detect invasive ductal carcinoma (IDC) from histopathology while remaining robust to the severe class imbalance characteristic of medical-imaging data.
- Designed a depth-wise separable CNN in TensorFlow/Keras for IDC classification on 50×50 histopathology patches, coupled with a reproducible preprocessing, class re-balancing, training, and evaluation pipeline.
- Demonstrated that the separable-convolution architecture matched standard convolutional baselines at substantially lower parameter count and computational cost, evidencing a favorable accuracy–efficiency trade-off on imbalanced data.
- First-author publication at ICAIE 2022.

PROFESSIONAL & APPLIED RESEARCH EXPERIENCE

Hughes Network Systems

June 2023 – Dec. 2025

Member of Technical Staff II, Software Engineering · Germantown, MD

- Framed satellite rain-fade prediction as a spatiotemporal learning problem and designed a deep 3D-CNN over highly imbalanced radar-imaging and telemetry time-series, achieving >90% classification accuracy with high precision and recall and improving downstream link reliability.
- Hypothesized that targeted synthetic augmentation would counter severe class imbalance; constructed customized synthetic time-series with tailored augmentation that improved F1 and recall by more than 5% and reduced performance variance across weather regimes.
- Developed an LSTM bandwidth-prediction model with a custom loss over streaming TCP-packet data, yielding a peer-reviewed publication (ICSSC 2024) and a pending U.S. patent.
- Built end-to-end ML pipelines (ingestion, automated quality validation, feature generation, real-time inference) and ran distributed multi-GPU / multi-node training (Docker, Kubernetes) to support large-scale experimentation.

NextTier

June 2024 – Sept. 2024

Software Engineer Intern — AI Reasoning Agents

- Investigated agentic LLM architectures for multi-step query resolution, building a multi-agent system with autonomous reasoning, dynamic tool use (retrieval, web search, code execution), and task decomposition, grounded by a retrieval-augmented generation (RAG) pipeline over domain knowledge.
- Designed an evaluation framework measuring task success rate, tool-call efficiency, failure-recovery rate, and multi-step completion accuracy, enabling systematic comparison of agent configurations and quantifying gains in factual grounding.

SELECTED PROJECTS

Residual Network — ResNet-50 Implemented from Scratch

[View Code ↗](#)

Personal Project · TensorFlow / Keras

- Reconstructed the ResNet-50 architecture (~23.6M parameters) from first principles—custom identity and projection residual blocks with skip connections and batch normalization—to probe how residual connections shape optimization and gradient flow in very deep networks.
- Trained and evaluated on a six-class hand-sign image-classification task, reaching ~87.5% test accuracy and empirically confirming the optimization advantage of residual learning over plain deep stacks.

Stock-Price Forecasting with Stacked LSTMs

[View Code ↗](#)

Personal Project · TensorFlow / Keras

- Studied sequence modeling for noisy financial time series, building a stacked four-layer LSTM (50→60→80→120 units) with dropout regularization to forecast next-step prices for Google (GOOG) and Oracle (ORCL) from multivariate OHLCV inputs.
- Engineered the sequence-windowing and normalization pipeline and trained per-ticker models under a mean-squared-error objective, comparing model variants to assess generalization across the two equities.

RELEVANT COURSEWORK

Machine Learning & AI: Machine Learning & Artificial Intelligence (A–)

Mathematics & Statistics: Linear Algebra (A), Real Analysis (A), Probability Theory (A), Linear Programming (A–), Matrices (A), Multivariable & Vector Calculus (A), Differential Equations (A–), Mathematical Statistics, Discrete Mathematics, Graph Theory

Computer Science: Programming Language Concepts (A–), Digital Logic Design (A–), Numerical Analysis I & II, Data Structures & Algorithms, Theory of Computation, Operating Systems, Systems Programming

TECHNICAL SKILLS

Languages: Python, C++, C, Java, MATLAB

ML / DL: PyTorch, TensorFlow, Keras, MMDetection, OpenCV, Scikit-learn, NumPy, Pandas; CNNs, 3D-CNNs, LSTMs; object detection (YOLO, YOLO-World), knowledge distillation, continual learning

Systems / Tools: Linux, Docker, Kubernetes, Git, FastAPI, PostgreSQL, Jenkins